STREPTOKINASE

Antigenic — once only

Hypotension during infusion

CHEMICAL NAME Bacterial protein (~47 kDa) produced by β-haemolytic group C streptococci

CLASS Non-fibrin-specific thrombolytic agent

TRADE NAMES Streptase · Kabikinase · STPase

THERAPEUTIC USES / INDICATIONS

- Acute STEMI (within 12 h) where newer agents unavailable (low cost).
- Massive pulmonary embolism; extensive DVT (rarely).
- Occluded arteriovenous shunts.

PHARMACODYNAMICS

- Forms a complex with **plasminogen** → activates other plasminogen to plasmin → systemic fibrinolysis (not clot-specific).
- Depletes fibrinogen.

ADMINISTRATION

- **IV infusion** — 1.5 million units in 100 mL saline over 60 min.
- Do not shake vial during reconstitution.
- May premedicate with hydrocortisone/chlorphenamine (allergy).
- Do not repeat if given >5 days previously (antibodies).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Hypotension during infusion (slow/stop and give fluids).
- Fever, chills, rash.
- Nausea.

ADVERSE EFFECTS (SERIOUS)

- Anaphylaxis and allergic reactions.
- Intracranial haemorrhage; major bleeding.
- Reperfusion arrhythmias.
- Guillain–Barré syndrome, vasculitis (rare).

DOSAGES

- **STEMI:** 1.5 million units IV over 60 min.
- **PE:** 250,000 units over 30 min, then 100,000 units/h for 24 h (or 1.5 MU over 2 h).

PHARMACOKINETICS

- **Onset:** immediate. Neutralised by anti-streptococcal antibodies.
- **t½:** ~23 min (plasminogen activator complex).

CONTRAINDICATIONS & CAUTIONS

- Previous streptokinase treatment (>5 days ago) or recent streptococcal infection (antibodies).
- Previous ICH; recent stroke; active bleeding; recent surgery/trauma.
- Suspected aortic dissection; severe hypertension.

KEY INTERACTIONS

- **Anticoagulants, antiplatelets:** ↑ bleeding.

NURSING MANAGEMENT

- Check history of previous streptokinase — document and give patient a card.
- BP every 15 min; slow or pause infusion if SBP <90.
- Monitor for allergic reaction, bleeding, arrhythmias.
- Assess chest pain and ST resolution.

PATIENT EDUCATION

- You must not receive streptokinase again; carry the card.
- Report bleeding, headache, rash or breathing difficulty.

KEY POINTS

- Non-fibrin-specific thrombolytic; cheap.
- Antigenic — never repeat after 5 days.
- 1.5 MU over 60 min for STEMI.
- Hypotension and allergy during infusion.